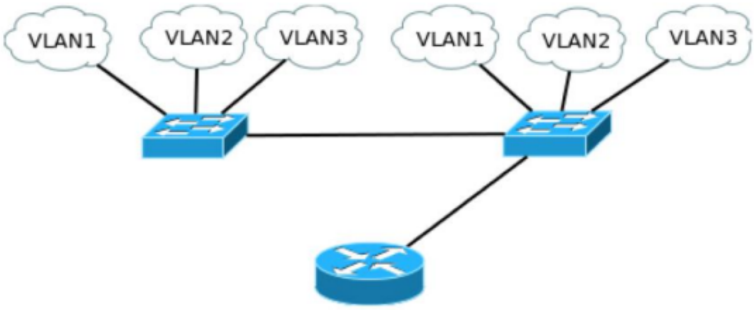
Guia 3– Inter-VLAN Routing

Objectivos:

- Definição de VLANs
- Routing entre VLANs

Inter-VLAN Routing: Router-on-a-Stick

3. Monte a rede apresentada na figura, onde foi adicionado um router.



- Configure o router para suportar sub-interfaces e routing Inter-VLAN (802.1Q):

```
Router(config)# interface FastEthernet0/0
Router(config-if)# no shutdown
Router(conficg-if)# interface FastEthernet0/0.1
Router(config-if)# encapsulation dot1Q 1 native          !VLAN1
Router(config-if)# ip address 10.1.1.1 255.255.255.0
!
Router(config-if)# interface FastEthernet0/0.2
Router(config-if)# encapsulation dot1Q 2        !VLAN2
Router(config-if)# ip address 10.2.2.1 255.255.255.0
!
Router(config-if)# interface FastEthernet0/0.3
Router(config-if)# encapsulation dot1Q 3        !VLAN3
Router(config-if)# ip address 10.3.3.1 255.255.255.0
```

Verifique a tabela de encaminhamento. Coloque terminais nas diferentes VLANs, configure neles a respetiva gateway (ou seja, os endereços IP das sub-interfaces do router) e teste a conetividade. Capture os pacotes trocados entre o router e o switch (da direita).

- **Dica1:** Vai precisar de ativar mais um trunk algures.
- **Dica2:** Se utilizou routers para substituir os hosts, não se esqueça de incluir a rota por defeito para fora da rede: ip route 0.0.0.0 0.0.0.0 *endereço_da_gateway*
- **Dica3:** Se tiver que reiniciar o GNS3 não se esqueça que tem que reativar a base de dados das vlans.

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5. Grave as configurações dos equipamentos, bem como o projecto.

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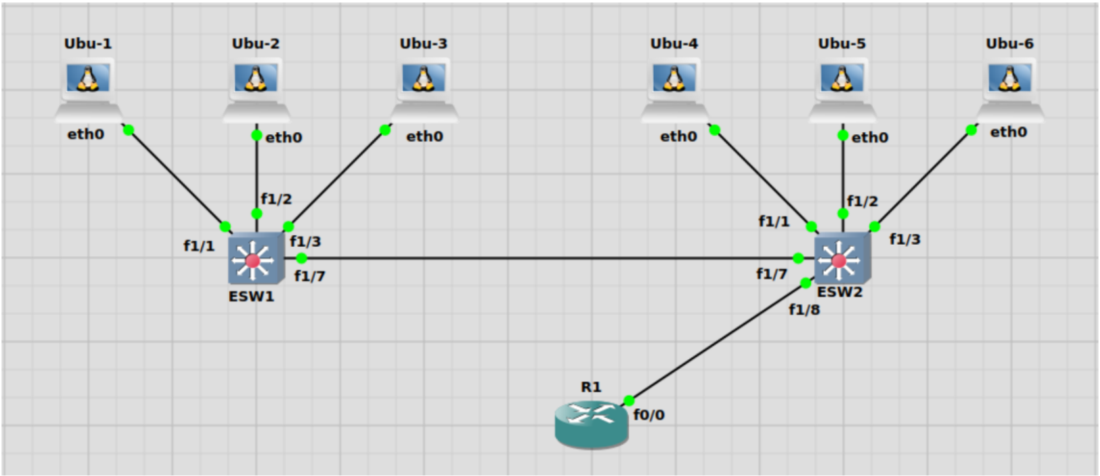

Ubu-1:

root@Ubu-1:~# cat /etc/network/interfaces

```
# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.1.1.2
        netmask 255.255.255.0
        gateway 10.1.1.1
```

Ubu-4:

root@Ubu-4:~# cat /etc/network/interfaces

```
# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.1.1.3
        netmask 255.255.255.0
        gateway 10.1.1.1
```

Ubu-2:

root@Ubu-2:~# cat /etc/network/interfaces

```
# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.2.2.2
        netmask 255.255.255.0
        gateway 10.2.2.1
```

Ubu-5:

root@Ubu-5:~# cat /etc/network/interfaces

```
# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.2.2.3
        netmask 255.255.255.0
        gateway 10.2.2.1
```

Ubu-3:

root@Ubu-3:~# cat /etc/network/interfaces

```
# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.3.3.2
        netmask 255.255.255.0
        gateway 10.3.3.1
```

Ubu-6:

root@Ubu-6:~# cat /etc/network/interfaces

```
# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.3.3.3
        netmask 255.255.255.0
        gateway 10.3.3.1
```

ESW1:

```
ESW1#vlan database
ESW1(vlan)#vlan 1
VLAN 1 modified:
ESW1(vlan)#vlan 2
VLAN 2 added:
    Name: VLAN0002
ESW1(vlan)#vlan 3
VLAN 3 added:
    Name: VLAN0003
ESW1(vlan)#exit
APPLY completed.
Exiting....
ESW1#sh vlan-switch

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa1/0, Fa1/1, Fa1/4, Fa1/5
                                                Fa1/6, Fa1/8, Fa1/9, Fa1/10
                                                Fa1/11, Fa1/12, Fa1/13, Fa1/14
                                                Fa1/15
2    VLAN0002                         active    Fa1/2
3    VLAN0003                         active    Fa1/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        1002   1003
2    enet  100002     1500  -      -      -        -    -        0      0
3    enet  100003     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        1      1003
1003 tr    101003     1500  1005   0      -        -    srb      1      1002
1004 fdnet 101004     1500  -      -      1        ibm  -        0      0
1005 trnet 101005     1500  -      -      1        ibm  -        0      0
```

```
ESW1(config-if)#int f1/7
ESW1(config-if)#shutdowm
                        ^
% Invalid input detected at '^' marker.

ESW1(config-if)#shutdow
ESW1(config-if)#shutdown
ESW1(config-if)#
*Mar  1 00:09:55.619: %LINK-5-CHANGED: Interface FastE
ratively down
*Mar  1 00:09:56.619: %LINEPROTO-5-UPDOWN: Line protoc
ged state to down
ESW1(config-if)#switchport mode trun
ESW1(config-if)#switchport mode trunk
ESW1(config-if)#switchport trunk enca
ESW1(config-if)#switchport trunk encapsulation dot
ESW1(config-if)#switchport trunk encapsulation dot1q
ESW1(config-if)#no shut
ESW1(config-if)#no shutdown
ESW1(config-if)#
```

ESW2:

```
ESW2#sh vlan-switch

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa1/0, Fa1/1, Fa1/4, Fa1/5
                                                Fa1/6, Fa1/9, Fa1/10, Fa1/11
                                                Fa1/12, Fa1/13, Fa1/14, Fa1/15
2    VLAN0002                         active    Fa1/2
3    VLAN0003                         active    Fa1/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        1002   1003
2    enet  100002     1500  -      -      -        -    -        0      0
3    enet  100003     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        1      1003
1003 tr    101003     1500  1005   0      -        -    srb      1      1002
1004 fdnet 101004     1500  -      -      1        ibm  -        0      0
1005 trnet 101005     1500  -      -      1        ibm  -        0      0
```

```
ESW2#sh interfaces trunk

Port      Mode         Encapsulation  Status        Native vlan
Fa1/7     on           802.1q         trunking      1
Fa1/8     on           802.1q         trunking      1

Port      Vlans allowed on trunk
Fa1/7     1-1005
Fa1/8     1-1005

Port      Vlans allowed and active in management domain
Fa1/7     1-3
Fa1/8     1-3

Port      Vlans in spanning tree forwarding state and not pruned
Fa1/7     1
Fa1/8     1
```

3.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#no shut
R1(config-if)#int
*Mar  1 00:01:32.851: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:01:33.851: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up
R1(config-if)#int f0/0.1
R1(config-subif)#enca
R1(config-subif)#encapsulation dot
R1(config-subif)#encapsulation dot1Q 1 native
R1(config-subif)#ip address
*Mar  1 00:02:05.131: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R1(config-subif)#ip address 10.1.1.1 255.255.255.0
R1(config-subif)#int f0/0.2
R1(config-subif)#encapsulation dot1Q 2
R1(config-subif)#ip address 10.2.2.1 255.255.255.0
R1(config-subif)#int f0/0.3
R1(config-subif)#encapsulation dot1Q 3
R1(config-subif)#ip address 10.3.3.1 255.255.255.0
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
C       10.3.3.0 is directly connected, FastEthernet0/0.3
C       10.2.2.0 is directly connected, FastEthernet0/0.2
C       10.1.1.0 is directly connected, FastEthernet0/0.1
```

```
Ubu-1 console is now available... Press RETURN to get started.
root@Ubu-1:~# ping 10.1.1.3
PING 10.1.1.3 (10.1.1.3) 56(84) bytes of data.
64 bytes from 10.1.1.3: icmp_seq=1 ttl=64 time=4.79 ms
64 bytes from 10.1.1.3: icmp_seq=2 ttl=64 time=0.565 ms
64 bytes from 10.1.1.3: icmp_seq=3 ttl=64 time=0.606 ms
^C
--- 10.1.1.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2036ms
rtt min/avg/max/mdev = 0.565/1.986/4.787/1.980 ms
root@Ubu-1:~# ping 10.2.2.3
PING 10.2.2.3 (10.2.2.3) 56(84) bytes of data.
64 bytes from 10.2.2.3: icmp_seq=2 ttl=63 time=18.1 ms
64 bytes from 10.2.2.3: icmp_seq=3 ttl=63 time=21.0 ms
64 bytes from 10.2.2.3: icmp_seq=4 ttl=63 time=21.3 ms
64 bytes from 10.2.2.3: icmp_seq=5 ttl=63 time=21.3 ms
^C
--- 10.2.2.3 ping statistics ---
5 packets transmitted, 4 received, 20% packet loss, time 4030ms
rtt min/avg/max/mdev = 18.126/20.441/21.345/1.345 ms
root@Ubu-1:~# ping 10.3.3.3
PING 10.3.3.3 (10.3.3.3) 56(84) bytes of data.
64 bytes from 10.3.3.3: icmp_seq=2 ttl=63 time=11.3 ms
64 bytes from 10.3.3.3: icmp_seq=3 ttl=63 time=16.3 ms
64 bytes from 10.3.3.3: icmp_seq=4 ttl=63 time=20.7 ms
^C
--- 10.3.3.3 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3016ms
rtt min/avg/max/mdev = 11.324/16.116/20.695/3.828 ms
```

Pela imagem anterior, é possível verificar que a máquina Ubu-1 tem conectividade com a Ubu-4 (e ambas estas máquinas encontram-se na VLAN 1), com a Ubu-5 (que se encontra na VLAN 2) e com a Ubu-6 (que se encontra na VLAN 3). Assim, verifica-se que existe conectividade entre as três VLANs.

Configuração do Router 1:

```
R1#sh run
Building configuration...

Current configuration : 1297 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
!
!
no ip domain lookup
ip auth-proxy max-nodata-conns 3
ip admission max-nodata-conns 3
!
!
!
!
!
!
!
```

```
!
!
!
!
!
!
!
!
!
!
ip tcp synwait-time 5
!
!
!
!
!
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface FastEthernet0/0.1
 encapsulation dot1Q 1 native
 ip address 10.1.1.1 255.255.255.0
!
interface FastEthernet0/0.2
 encapsulation dot1Q 2
 ip address 10.2.2.1 255.255.255.0
!
interface FastEthernet0/0.3
 encapsulation dot1Q 3
 ip address 10.3.3.1 255.255.255.0
!
interface FastEthernet0/1
 no ip address
 shutdown
 duplex auto
 speed auto
!
interface FastEthernet1/0
 no ip address
 shutdown
 duplex auto
 speed auto
!
```

```
interface FastEthernet2/0
 no ip address
 shutdown
 duplex auto
 speed auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
end
```

```
R1#sh ip int br
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            unassigned      YES unset  up                    up
FastEthernet0/0.1          10.1.1.1        YES manual up                    up
FastEthernet0/0.2          10.2.2.1        YES manual up                    up
FastEthernet0/0.3          10.3.3.1        YES manual up                    up
FastEthernet0/1            unassigned      YES unset  administratively down down
FastEthernet1/0            unassigned      YES unset  administratively down down
FastEthernet2/0            unassigned      YES unset  administratively down down
```

Questões finais:

a) Se a mesma rede IP dentro das VLANs, consigo manter conectividade entre estações da mesma VLAN? Se as estações estiverem na mesma VLAN e na mesma rede, sim é possível, pois o tráfego entre a mesma VLAN é encaminhado diretamente pelo trunk, não vai através do router.

b) Se a mesma rede IP dentro das VLANs, consigo manter conectividade entre estações de VLANs diferentes?
Não, porque o Router que vai encaminhar o tráfego entre diferentes VLANs, para cada subinterface, só pode ter IPs de redes diferentes configurados em cada subinterface. E sendo que as estações estão em VLANs diferentes, é necessário criar uma nova subinterface numa rede diferente, na qual não é possível configurar o endereço IP na mesma rede. No entanto, se as redes fossem diferentes, seria possível.